

The Metric side of Optimal Transport

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Outline

The Disintegration Theorem for Product Spaces

A special case of the Disintegration Theorem we need is the Following.
Throughout the talk we assume that all spaces are Polish.

Theorem

Let X, Y be Polish, $p : X \times Y \rightarrow X$ the projection and $\pi \in \mathcal{P}(X \times Y)$.
Setting $\mu := p_{\#}(\pi)$ we get the existence of a parametrized family of probability measures $\{\pi_x\}_{x \in X} \subset \mathcal{P}(X \times Y)$ such that

1. for all $A \in \mathcal{B}(X \times Y)$ the function $x \mapsto \pi_x(A)$ is measurable.
2. $\pi(A) = \int_X \pi_x(A) d\mu(x)$ for all $A \in \mathcal{B}(X \times Y)$
3. π_x lives on $p^{-1}(x) = \{x\} \times Y$ for $x \in X$.

The Dudley Lemma

In the Literature this is often called Gluing Lemma!

Lemma

Let $(X_1, \mu_1), (X_1, \mu_2), (X_1, \mu_3)$ be Polish and $\pi^{1,2} \in \Gamma(\mu_1, \mu_2)$ and $\pi^{2,3} \in \Gamma(\mu_2, \mu_3)$. Then there exists some $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ such that

$$p_{\#}^{1,2}(\pi) = \pi^{1,2} \quad \text{and} \quad p_{\#}^{2,3}(\pi) = \pi^{2,3}$$

where $p^{1,2}(x_1, x_2, x_3) = (x_1, x_2)$ and $p^{2,3}(x_1, x_2, x_3) = (x_2, x_3)$.

Proof is essentially just taking the product measures of the Disintegration's of $\pi^{2,3}$ and $\pi^{1,2}$ and "Gluing" them together with μ_2 .

The Wasserstein distance for finite p

As a reminder, we have for any $p \in [1, \infty[$

$$\mathcal{P}_p(X) := \left\{ \mu \in \mathcal{P}(X) \mid \int_X d^p(x, x_0) d\mu(x) < \infty \right\}$$

and the following will be a metric on this space.

Definition

For any $p \in [1, \infty[$ and $\nu, \mu \in \mathcal{P}_2(X)$

$$W_p^p(\mu, \nu) := \min \left\{ \int_{X \times X} d^p(x, y) d\pi(x, y) \mid \pi \in \Gamma(\mu, \nu) \right\}.$$

Theorem

$(\mathcal{P}_p(X), W_p)$ is a metric space!

What do we already know about this distance?

Firstly, we get that

$x \mapsto \delta_x$ is an isometry.

And, by the Kantorovich-Rubinstein Duality we have

$$\begin{aligned} W_1(\mu, \nu) &= \sup_{(\phi, \psi) \in I_c} \left\{ \int_X \phi d\mu + \int_X \psi d\nu \right\} \\ &= \sup_{\|\phi\|_{Lip} \leq 1} \left\{ \int_X \phi d\mu + \int_X \phi^c d\nu \right\} \\ &= \sup_{\|\phi\|_{Lip} \leq 1} \left\{ \int_X \phi d\mu - \int_X \phi d\nu \right\} \end{aligned}$$

Comparing L^2 norm and Wasserstein distance

Let $\nu \ll \lambda^n$ with Radon Nikodym derivative f such that $\text{supp}(f) \subseteq \overline{B_1}$.

Additionally, let $\mu_h \ll \lambda^n$ such that $f_h(x) = f(x + h)$. Then we have

$$\|f_h - f\|_{L^2(\mathbb{R}^n)}^2 = 2\|f\|_{L^2(\mathbb{R}^n)}^2$$

but on the other hand

$$W_2(\mu_h, \mu) = \|h\|_2$$

which for large $\|h\|$ implies

$$W_2(\mu_h, \mu) \gg \|f_h - f\|_{L^2(\mathbb{R}^n)}.$$

In case of small $\|h\|$ we can also find $W_2(\mu_h, \mu) \ll \|f_h - f\|_{L^2(\mathbb{R}^n)}$.

Wasserstein distance for normal distributions

For any $\mu_X, \mu_Y \in \mathcal{P}_2(\mathbb{R}^d)$ with $\mathcal{L}(X) = \mu_X$ and $\mathcal{L}(Y) = \mu_Y$ where X and Y are normally distributed, the Wasserstein distance can be calculated as follows

$$W_2(\mu_X, \mu_Y) = \|m_X - m_Y\|^2 + \text{tr} \left(\Sigma_X - 2 \cdot \left(\Sigma_Y^{1/2} \Sigma_X \Sigma_Y^{1/2} \right)^{1/2} + \Sigma_Y \right)$$

If Σ_X and Σ_Y commute, this simplifies to

$$W_2(\mu_X, \mu_Y) = \|m_X - m_Y\|^2 + \sum_{i=1}^d \left(\sqrt{\lambda_i^X} - \sqrt{\lambda_i^Y} \right)^2$$

where λ_i^X and λ_i^Y are the Eigenvalues of Σ_X and Σ_Y respectively.

Normally distributed examples

Wasserstein distance for $p = \infty$

If we restrict ourselves to $\mathcal{P}_\infty(X)$, the space of probability measures with bounded support. We can also define W_∞ as the limit of W_p .

$$\begin{aligned}\lim_{p \rightarrow \infty} W_p(\mu, \nu) &= W_\infty(\mu, \nu) = \inf\{\|d(x, y)\|_{L^\infty}(\pi) \mid \pi \in \Gamma(\mu, \nu)\} \\ &= \inf_{\pi \in \Gamma(\mu, \nu)} \inf\{C \geq 0 \mid |d(x, y)| \leq C \text{ for } \pi \text{ almost all } (x, y)\}\end{aligned}$$

and we have

$(\mathcal{P}_\infty(X), W_\infty)$ is a metric space

Iterated Dudley Lemma

We want to represent a sequence of measures (μ_n) as the marginals of some measure in an infinite product space.

For that we need the iterated Dudley Lemma. Note that we also allow $N = \infty$ in this Lemma, in that case, the $\leq$ become $<$.

Lemma

Let $N \geq 3$ and for any $n \leq N$ (X_n, d_n) Polish, $\mu_n \in \mathcal{P}(X_n)$ and $\theta_n \in \Gamma(\mu_{n-1}, \mu_n)$. Then there exists $\pi_n \in \mathcal{P}(X_1 \times \dots \times X_n)$ for any $n \leq N$ such that.

1. $p_{\#}^{1, \dots, n-1} \pi_n = \pi_{n-1}$ for $2 \leq n \leq N$
2. $p_{\#}^i \pi_n = \mu_i$ for $1 \leq i \leq n \leq N$
3. $p_{\#}^{i-1, i} \pi_n = \theta_i$ for $2 \leq i \leq n \leq N$

Proof is just applying the Dudley Lemma iteratively (hence the name :).

Lifting completeness from X to $\mathcal{P}_p(X)$

For $N = \infty$ we get a sequence (π_n) and can find a unique measure π_∞ on $\mathbb{X} = \prod_{i=1}^{\infty} X_i$ such that

$$p_{\#}^{(1, \dots, n)} \pi_\infty = \pi_n.$$

Additionally, we remind ourselves of the metric version of the L^p spaces.

$$L^p(\Omega, \mathcal{F}, P, X) := \left\{ f : \Omega \rightarrow X \mid f \text{ measurable, } \int_{\Omega} d^p(f, z_0) dP < \infty \right\}$$

with

$$d_{L^p}^p(f, g) := \int_{\Omega} d^p(f, g) dP$$

These are not necessarily vector spaces, but one can show that this more general notion of L^p space is still complete. (Which we will need in the following)

Theorem

Let (X, d) be a complete metric space, then $(\mathcal{P}_p(X), W_p)$ is complete.